

ECE 447

Lesson 27

Digital Carrier Modulation and Demodulation (Bandpass Signals)

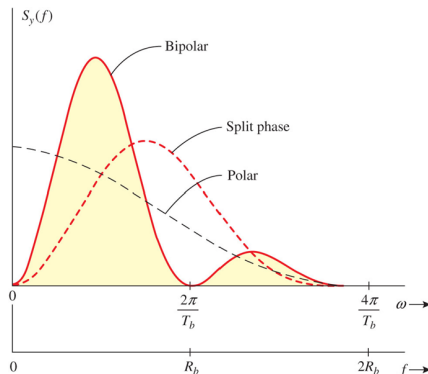


UNITED STATES
AIR FORCE
ACADEMY

SCHEDULE AND ADMIN

- [Schedule](#).
- Admin
 - **HW4**. Graded. Submit any regrade requests via Gradescope.
 - **Lab 5**. Graded. Submit any regrade requests via Gradescope.
 - **HW5**. Due tonight (Lesson 27, 23 Oct).
 - **HW6**. Assigned today. Due Lesson 30 (Mon, 2 Nov).
 - **Lab 6**. Begins Lesson 28 (27 Oct); PDF due Lesson 31 (Wed, 4 Nov) to Gradescope.

REVIEW



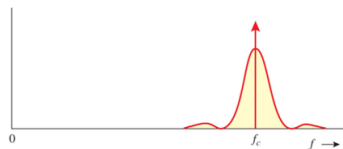
- The baseband polar signaling BW in this figure is for RZ. If changed to NRZ (full-width pulse), the bandwidth reduces to R_b . (Important for HW 6 6.8-1)

BINARY CARRIER MODULATIONS

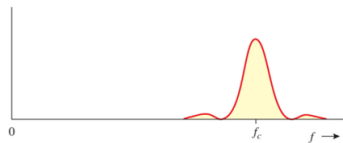
- ASK - self-explanatory
- Angle modulations: PSK and FSK
- PSK
 - If $m(t)$ uses polar signaling $\{\pm 1\}$, phase shifts for BPSK signal, $\varphi_{PSK}(t)$, is π radians apart
 - $\varphi_{PSK}(t) = m(t) \cos(\omega_c t)$
 - Constellation plot?
- FSK:
 - Sum of two alternating ASK signals with different carrier frequencies, ω_{c0} and ω_{c1}
 - $a_k = \{0, 1\}$
 - $\varphi_{FSK}(t) = \sum a_k p(t - kT_b) \cos(\omega_{c1} t) + \sum (1 - a_k) p(t - kT_b) \cos(\omega_{c0} t)$
- FT modulation property derives PSDs of digital bandpass signals:

$$S_{\varphi}(f) = \frac{1}{4} S_M(f + f_c) + \frac{1}{4} S_M(f - f_c)$$

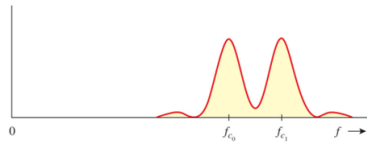
BINARY CARRIER MODULATIONS



(a)



(b)



(c)

M-ARY CARRIER MODULATIONS

- Focus on QAM - most widely used in modern systems

